# Soil Health and Compost in Organic Farming

Test document 2 of 5 · Soil organic matter, compost, and nutrient cycling

Purpose: This PDF is an original, concise test document designed to exercise multi-PDF retrieval and page-level citations. It summarizes general concepts and points to authoritative public references; it is not a replacement for official standards or local agricultural advice.

## 1. Why soil health matters

Healthy soil supports roots, stores and releases water, cycles nutrients, and provides habitat for microorganisms and soil fauna. Soil health is influenced by physical structure, chemical properties, and biological activity; no single measurement describes every aspect of it.

Organic matter can improve aggregation and water retention, while living roots and diverse soil organisms support nutrient cycling. Results depend on soil type, climate, crop, and management.

## 2. Compost basics

Composting is the managed biological decomposition of suitable organic materials. A practical process balances carbon-rich materials, nitrogen-rich materials, moisture, oxygen, and time. A pile that is too wet may become oxygen-poor; a pile that is too dry may decompose slowly.

Use mature, stable compost and follow local food-safety and organic-standard requirements. Poorly handled manure or immature compost can create crop, nutrient, or contamination problems.

## 3. Other soil-building practices

Cover crops protect bare soil, add biomass, and can capture nutrients that might otherwise be lost. Legume cover crops may contribute biologically fixed nitrogen when compatible with the system.

Crop residues and mulches reduce the impact of raindrops and can limit evaporation. Reduced tillage may protect soil structure, but the best approach depends on weed pressure, equipment, soil drainage, and crop needs.

Use soil tests and field observations to decide whether amendments are needed. More compost is not always better: repeated heavy applications can accumulate phosphorus or salts.

## 4. Practical checklist

Record material sources and approximate quantities.

Keep composting areas away from watercourses and manage runoff.

Monitor moisture and aeration during composting.

Test soil periodically and match nutrient additions to crop needs.

Combine organic amendments with crop rotation, cover crops, and erosion control.

## Questions to test your RAG application

1. How can compost improve soil, and what risks can occur with excessive applications?
2. What conditions can make a compost pile decompose poorly?
3. How do cover crops and crop residues help protect soil?

## Public references

**ICAR / MANAGE — Soil Health Management in Organic Farming**
https://www.manage.gov.in/publications/eBooks/organic%20farming.pdf

**USDA NRCS — National Organic Farming Handbook**
https://www.nrcs.usda.gov/sites/default/files/2022-12/Organic_Farming_Handbook2.pdf

**USDA AMS — Organic Standards Tip Sheets**
https://www.ams.usda.gov/reports/tip-sheets-organic-standards

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